

GLOSSARY

A

absolute magnitude see *magnitude*.

absorption line see *spectral line*.

absorption nebula see *nebula*.

accelerating universe A universe that expands at an accelerating rate. Current evidence indicates that the expansion of our universe had been slowing down under the action of gravity until about 6 billion years ago, but that since then it has been accelerating. The acceleration is believed to be driven by the repulsive influence of dark energy. See also *dark energy*.

accretion (1) The colliding and sticking together of small, solid particles and bodies to make progressively larger ones. **(2)** The process by which a body grows in mass by accumulating matter from its surroundings. An **accretion disk** is a disk of gas that revolves around a star or a compact object such as a white dwarf, neutron star, or black hole and has been drawn in from a companion star or from neighboring gas clouds.

active galaxy A galaxy that emits an exceptional amount of energy over a wide range of wavelengths, from radio waves to X-rays. An **active galactic nucleus (AGN)** is the compact, highly luminous core of an active galaxy that, in many cases, varies markedly in brightness over time, and is thought to be powered by the accretion of gas onto a supermassive black hole. See also *black hole*, *galaxy*.

active prominence see *prominence*.

albedo The ratio of the amount of light reflected by a body, such as a planet or a part of a planet's surface, to the amount of light that it receives from the Sun. Albedo values range from 0, for a perfectly dark object that reflects nothing, to 1, for a perfect reflector.

altazimuth mounting A mounting that enables a telescope to be rotated in altitude (around a horizontal axis) and in azimuth (around a vertical axis). Many large modern telescopes are mounted in this way, using computer-controlled motors to drive the telescope in altitude and azimuth so as to track the motion of an object across the sky. See also *altitude*, *azimuth*, *equatorial mounting*.

altitude The angular distance between the horizon and a celestial body. Altitude takes values from 0° (for an object on the horizon) to 90° (for an object that is directly overhead). See also *azimuth*.

annual parallax see *parallax*.

annual proper motion see *proper motion*.

annular eclipse see *eclipse*.

antimatter Material composed of antiparticles. See *antiparticle*.

antiparticle An elementary particle that has the same mass as a particle of ordinary matter but exactly opposite values of other quantities such as spin and electrical charge. For example, the antiparticle of the negatively charged electron is the positively charged **positron**. If a particle and its antiparticle collide, both are annihilated and converted into energy.

aperture The clear diameter of the objective lens or primary mirror of a telescope or other optical instrument.

aphelion The point on its elliptical orbit at which a body such as a planet, asteroid, or comet is at its greatest distance from the Sun.

apogee The point on its elliptical orbit around Earth at which a body such as the Moon or a spacecraft is at its greatest distance from Earth. See also *perigee*.

apparent magnitude see *magnitude*.

arachnoid A type of structure, found on the surface of Venus, that consists of concentric circular or oval fractures or ridges, together with a complex network of fractures or ridges that radiate outward. Its name derives from its superficial resemblance to a spiderweb. Typical diameters range from 30–110 miles (50 to 175 km).

asterism A conspicuous pattern of stars that is not itself a constellation. A well-known example is the Big Dipper, which forms part of the constellation Ursa Major (the Great Bear). See also *constellation*.

asteroid One of the vast number of small bodies that revolve independently around the Sun. Their diameters range from a few yards (meters) to around 600 miles (1,000 km). While the greatest concentration of asteroids is in the **Main Belt**, which lies between the orbits of Mars and Jupiter, asteroids are found throughout the solar system. A **near-Earth asteroid (NEA)** is a body whose orbit comes close to, or intersects, the orbit of Earth. Formally, a near-Earth asteroid is defined as one that has a perihelion distance of less than 1.3 times Earth's mean distance from the Sun. See also *Kuiper Belt*.

astronomical unit (au) A unit of distance measurement equal to the semimajor axis of Earth's elliptical orbit, equivalent to the average of the maximum and minimum distances between Earth and Sun. 1 au = 92,956,000 miles (149,598,000 km).

atom A basic building block of matter that is the smallest unit of a chemical element possessing the characteristics of that element. It consists of a nucleus of protons and neutrons, surrounded by a cloud of electrons. An atom has the same number of orbiting electrons as it has protons, so it is neutral (has no electrical charge). The chemical identity of an atom is determined by the number of protons in its nucleus (its **atomic number**). An atom of hydrogen (the simplest and lightest element) consists of a single proton and a single electron. See also *electron*, *neutron*, *proton*.

aurora A glowing, fluctuating display of light that is produced when charged particles entering a planet's upper atmosphere, usually in the vicinity of its north and south magnetic poles, collide with atoms and stimulate them to emit light.

autumnal equinox see *equinox*.

azimuth The angle between the north point on an observer's horizon and a celestial object, measured in a clockwise direction around the horizon. The azimuth of due north is 0°, due east 90°, due south 180°, and due west 270°. See also *altitude*.

B

background radiation see *cosmic microwave background radiation*.

barred spiral galaxy A galaxy that has spiral arms emanating from the ends of an elongated, bar-shaped, nucleus. See also *galaxy*, *spiral galaxy*.

baryon A particle, composed of three quarks, that is acted on by the strong nuclear force. Examples include protons and neutrons, the building blocks of atomic nuclei.

Big Bang The event in which the universe was born. According to Big Bang theory, the universe originated a finite time ago in an extremely hot, dense initial state and ever since then has been expanding. The Big Bang was the origin of space, time, and matter.

Big Crunch The final state that will be reached by the universe if it eventually ceases to expand and then collapses in on itself.

Big Rip The tearing apart of all forms of structure in the universe—galaxy clusters, galaxies, stars, planets, atoms, and elementary particles—that is expected to occur should the repulsive effect of dark energy become infinitely strong in a finite time. See also *dark energy*.

binary star Two stars that revolve around each other under the influence of their mutual gravitational attraction.

Each member star orbits the center of mass of the system, a point that lies closer to the more massive of the two stars. A **spectroscopic binary** is a system in which the two stars are too close to be resolved into separate points of light, but whose binary nature is revealed by its spectrum. The combined spectrum of the two stars contains two sets of spectral lines that shift in wavelength as the stars revolve around each other. An **eclipsing binary** is a system in which each star alternately passes in front of the other, cutting off all or part of its light and causing a periodic variation in the combined light of the two stars. See also *Doppler effect*, *spectral line*.

BL Lacertae object A type of active galaxy that has no detectable absorption or emission lines in its spectrum but which is believed to be similar to a quasar. The name derives from an object in the constellation Lacerta that was at first thought to be a variable star. See also *quasar*.

black body An idealized body that absorbs and reemits all the radiation that falls on its surface and which is a perfect radiator. A black body emits a continuous spectrum of radiation (**black-body radiation**) that peaks in brightness at a wavelength that depends on its surface temperature—the higher the temperature, the shorter the wavelength of peak brightness. See also *spectrum*.

black dwarf star A white dwarf star that has cooled to such a low temperature that it emits no detectable light. There has not been enough time since the origin of the universe for any star to cool down enough to become a black dwarf. See also *brown dwarf star*, *white dwarf star*.

black hole A compact region of space, surrounding a collapsed mass, within which gravity is so powerful that no material object, light, or any other kind of radiation can escape to the outside universe. The radius of a black hole is called the **Schwarzschild radius**, and its boundary is known as the **event horizon**. The greater its mass, the larger its radius. When a body collapses to form a black hole, all of its mass becomes compressed into a central point, a point of infinite density called a **singularity**. A **stellar-mass black hole** forms when the core of a high-mass star collapses; its mass is likely to be in the region of 3–100 times the mass of the Sun. A **super-massive black hole**, with a mass in the region of a few million to a few billion solar masses, is an object that forms when a very large mass collapses, or a number of black holes merge into one, in the core of a galaxy. See also *active galaxy*, *singularity*.